

Gursevak Singh Aulakh

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PROFESSIONAL SUMMARY

Final-year Computer Engineering student with hands-on experience in **Artificial Intelligence, Android Development, Computer Vision, IoT, and Full-Stack Web Development**. Co-inventor of a **published Indian patent** on AI-powered missing person detection and family reunification for mass gatherings using hybrid identity fusion. Passionate about building scalable, real-world AI systems that integrate machine learning, embedded hardware, and cloud technologies to solve impactful societal problems.

EDUCATION

Guru Gobind Singh College of Engineering & Research Centre, Nashik

2023 – 2026

Bachelor of Engineering in Computer Engineering

Savitribai Phule Pune University

– **CGPA: 8.46/10**

Guru Gobind Singh Polytechnic, Nashik

2020 – 2023

Diploma in Computer Engineering

Maharashtra State Board of Technical Education

– **Percentage: 86.57%**

Guru Gobind Singh Public School, Nashik

2019 – 2020

Secondary School Certificate (SSC)

Maharashtra State Board

– **Percentage: 84.40%**

PROJECTS

Kumbh Bandhu: Snan to Suraksha | Kotlin, Firebase, TensorFlow Lite, ArcFace CNN, ESP32, RFID, React

- Architected an AI-powered missing person detection and family reunification platform for large-scale public gatherings, integrating Android, Edge AI, RFID, and cloud technologies; technology protected under a **published Indian patent (Application No. 202621047713 A)**.
- Developed a real-time facial recognition pipeline using **ArcFace (512-dimensional embeddings)** with TensorFlow Lite, achieving **94.7% identification accuracy** under real-world conditions.
- Designed a hybrid identity fusion framework combining facial embeddings, RFID observations, gait signatures, and behavioral traits into a composite confidence score for reliable identification and reduced false positives.
- Implemented a complete Android-based volunteer coordination workflow with live RFID tracking, Firebase synchronization, QR verification, and location-aware emergency response, reducing manual search effort by approximately **73%**.
- Awarded **2nd Prize at National Level Techno Fest 2026** and shortlisted among the **Top 33 of 135+ teams** at Synergy 2026 conducted by The Institution of Engineers (India).

EVA: AI Institutional Desk Robot | Python, FastAPI, React, SQLite, ESP32-S3, Gemini AI, Kotlin, WebSocket

- Engineered an AI-powered cyber-physical institutional assistant that integrates conversational AI, robotics, web services, and embedded systems to automate academic, administrative, and student support operations within educational departments.
- Designed a modular microservice architecture using **FastAPI, React, SQLite, and Gemini AI**, enabling natural language querying across student records, attendance, faculty schedules, timetables, and departmental data through intelligent tool orchestration.
- Developed a physical robotic interface powered by **ESP32-S3** featuring voice interaction, dual OLED expressive eyes, I2S audio pipeline, and real-time communication with backend AI services, creating an interactive HOD desk assistant.
- Built a secure administrative dashboard with bulk Excel import, CRUD management, attendance analytics, AI-powered departmental insights, and live synchronization between the web platform, robot, and Android companion application.
- Implemented an Android companion application using **Kotlin** and WebSockets to execute AI-triggered actions including intelligent call initiation, enabling seamless coordination between the robot, backend services, and departmental staff.

- Designed and delivered the complete end-to-end system—including software architecture, AI integration, backend APIs, embedded firmware, database design, and hardware prototype—within a few weeks as a production-oriented institutional automation platform.

Granthalya – Sikh Granth Digital Library | *React, TypeScript, Supabase, PostgreSQL, TanStack Query*

[Live Demo](#)

- Developed a full-stack digital library platform preserving Sikh literature with verse-wise reading, multilingual translations, and structured navigation across more than **218 chapters**.
- Built a Scholar's Workbench featuring a triple-column verse editor, editorial workflow (Draft → Review → Published), audio timestamp synchronization, and secure role-based access using Supabase Authentication.
- Designed a scalable hierarchical database architecture with PostgreSQL and optimized client-side caching using TanStack Query, enabling efficient retrieval of thousands of verses and metadata records.

BhashaScan – Indian Language OCR & Translator | *Python, OpenCV, Tesseract OCR, Streamlit*

- Built an OCR and translation platform supporting text extraction from images and PDFs across **9 Indian languages** with automatic multilingual translation.
- Engineered an adaptive preprocessing pipeline using Laplacian variance-based quality estimation and OpenCV image enhancement, improving OCR accuracy by approximately **40%** on noisy scanned documents.
- Implemented multi-page PDF processing, 300 DPI rendering, lazy loading, batch translation, and downloadable text export through an interactive Streamlit interface.

PATENT

Published Indian Patent – AI-Powered Missing Person Detection & Family Reunification System

Published Jun 2026

Co-Inventor (1 of 7) | Indian Patent Application No. 202621047713 A

Filed Apr 2026

- Co-invented a **published Indian patent** introducing a privacy-preserving, multi-modal identity recognition framework for locating missing persons and enabling family reunification in large-scale public gatherings.
- Designed a **hybrid identity fusion architecture** that combines facial embeddings, RFID observations, gait signatures, and behavioral traits into a unified confidence score, significantly improving identification reliability.
- The patented architecture serves as the technical foundation of the **Kumbh Bandhu** platform, integrating Edge AI, Android, IoT, and cloud technologies for real-time emergency response.

EXPERIENCE

Data Science Intern

Jan 2025 – Mar 2025

AcmeGrade (Remote)

- Performed data preprocessing, feature engineering, and exploratory data analysis (EDA) using Python, Pandas, and NumPy to prepare datasets for machine learning workflows.
- Built and evaluated machine learning pipelines using Scikit-learn, applying data visualization and statistical analysis to derive actionable insights.
- Presented analytical findings through visual dashboards and reports, improving interpretation of model performance and business metrics.

Web Development Trainee

Jul 2022 – Aug 2022

Calibers Infotech, Nashik

- Developed dynamic web applications using PHP, MySQL, HTML, CSS, and JavaScript following MVC-based development practices.
- Implemented user authentication, database connectivity, server-side form handling, and CRUD operations for responsive web applications.
- Collaborated with senior developers to debug applications, optimize SQL queries, and improve overall application performance.

Workshop Facilitator – AI Tools for Engineers

Jun 2026

Department of Computer Engineering, Guru Gobind Singh Polytechnic, Nashik

- Conducted a two-day hands-on workshop for students and faculty on Generative AI, Prompt Engineering, AI-assisted software development, and productivity tools.
- Delivered live demonstrations using ChatGPT, Claude, Gemini, Perplexity, NotebookLM, Gamma, and Lovable for coding, research, documentation, presentation creation, and rapid application development.
- Mentored participants in building AI-powered workflows and practical projects, enabling effective adoption of modern AI tools in academic and engineering applications.

TECHNICAL SKILLS

Programming Languages: Java, Kotlin, Python, C++, JavaScript, TypeScript, PHP, SQL

Mobile Development: Android SDK, Jetpack Components, Firebase Authentication, Firebase Realtime Database, Firebase Storage, Cloud Functions

Web Development: React, Vite, HTML5, CSS3, JavaScript (ES6+), TypeScript, PHP, REST APIs

Artificial Intelligence & Machine Learning: TensorFlow Lite, ArcFace, Computer Vision, OpenCV, Tesseract OCR, Scikit-learn, Pandas, NumPy, Feature Engineering, Exploratory Data Analysis (EDA)

Databases & Cloud: Firebase, PostgreSQL, MySQL, Supabase

Embedded Systems & IoT: ESP32, RFID (HF/UHF), Edge AI, Real-Time Device Communication

Developer Tools: Git, GitHub, Android Studio, VS Code, Jupyter Notebook

Core Computer Science: Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, RESTful APIs

ACHIEVEMENTS & LEADERSHIP

Published Indian Patent – Co-Inventor

Jun 2026

Indian Patent Application No. 202621047713 A

- Co-invented a published patent on AI-powered missing person detection and family reunification using hybrid identity fusion, Edge AI, RFID, and computer vision.

National Level Techno Fest 2026 – Second Prize

Apr 2026

L.G.N. Sapkal College of Engineering, Nashik

- Awarded **2nd Prize** for developing the AI-powered **Kumbh Bandhu** platform, competing against engineering teams from multiple institutions.

Synergy 2026 – Top 33 Finalist

Apr 2026

The Institution of Engineers (India), Nashik Local Centre

- Selected among the **Top 33 teams out of 135+** entries for presenting an innovative AI-enabled crowd safety solution for mass gatherings.

AI Workshop Facilitator

Jun 2026

Guru Gobind Singh Polytechnic, Nashik

- Conducted a two-day hands-on workshop on Generative AI, Prompt Engineering, and AI-assisted software development for students and faculty.

Research Paper Presentation

Jun 2023

ICRTETM 2023

- Presented the paper titled "*Animal Rescue and Wellness Web Application*" at the Second International Conference on Recent Trends in Engineering, Science, Technology and Management.

Telecom Technician – IoT Devices/Systems (Level 4)

Sept 2023

Telecom Sector Skill Council (TSSC)

Grade: B

- Successfully completed the NSQF-certified IoT Devices/Systems training program recognized by NCVET.

ADDITIONAL INFORMATION

Languages: English (Professional), Hindi (Native), Punjabi (Native)

Areas of Interest: Artificial Intelligence, Computer Vision, Android Development, IoT, Embedded Systems, Full-Stack Development

Soft Skills: Leadership, Public Speaking, Technical Mentoring, Problem Solving, Team Collaboration